Lesson 26

Solve each rational inequality and graph the solution set.

1) $(x-2)(x+3) \le 0$

a) Graph the function $y=(x-2)(x+3)$

b) The x-intercepts of the function $y=(x-2)(x+3)$ are ___?___

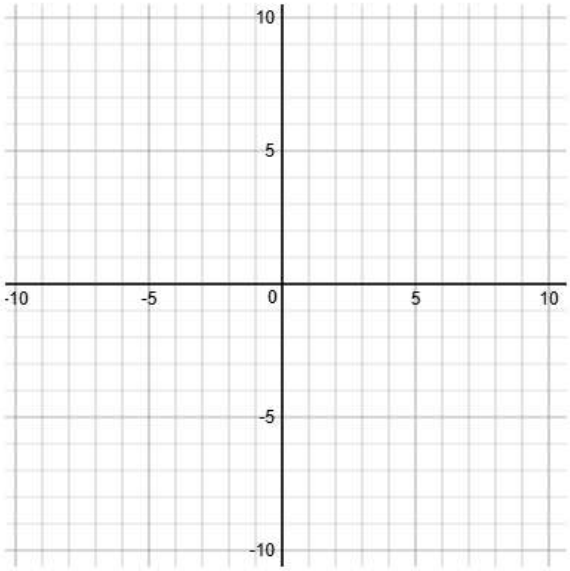
c) The interval on which the graph $y=(x-2)(x+3)$ is below the x-axis is _____?_____

d) The solution set for the inequality $(x-2)(x+3) \le 0$ is ___?___

e) Is $x=-5$ a solution to the inequality $(x-2)(x+3) \le 0$?

f) Is $x=0$ a solution to the inequality $(x-2)(x+3) \le 0$?

g) Is $x=1$ a solution to the inequality $(x-2)(x+3) \le 0$?



Solve each rational inequality and graph the solution set.

2) $(x-4)(x+2) > 0$

a) Graph the function $y=(x-4)(x+2)$

b) The x-intercepts of the function $y=(x-4)(x+2)$ are ___?___

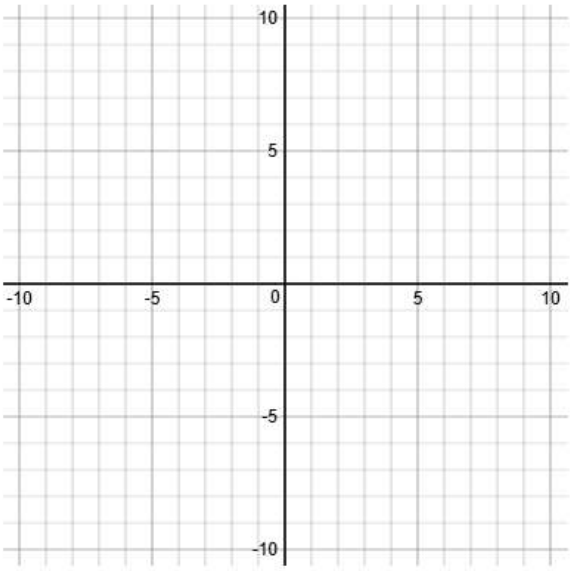
c) The interval on which the graph $y=(x-4)(x+2)$ is above the x-axis is _____?_____

d) The solution set for the inequality $(x-4)(x+2) > 0$ is ___?___

e) Is $x=-4$ a solution to the inequality $(x-4)(x+2) > 0$?

f) Is $x=0$ a solution to the inequality $(x-4)(x+2) > 0$?

g) Is $x=5$ a solution to the inequality $(x-4)(x+2) > 0$?



Solve each rational inequality and graph the solution set.

3) $x^2+4x+3 \ge 0$

a) Graph the function $y=x^2+4x+3$

b) The x-intercepts of the function $y=x^2+4x+3$ are _____?_____

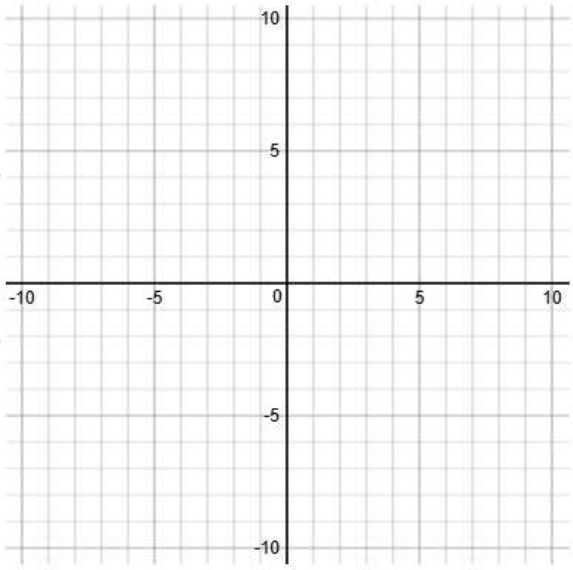
c) The interval on which the graph $y=x^2+4x+3$ is above the x-axis is _____?_____

d) The solution set for the inequality $x^2+4x+3 \ge 0$ is _____?_____

e) Is $x=-7$ a solution to the inequality $x^2+4x+3 \ge 0$?

f) Is $x=1$ a solution to the inequality $x^2+4x+3 \ge 0$?

g) Is $x=6$ a solution to the inequality $x^2+4x+3 \ge 0$?



Solve each rational inequality and graph the solution set.

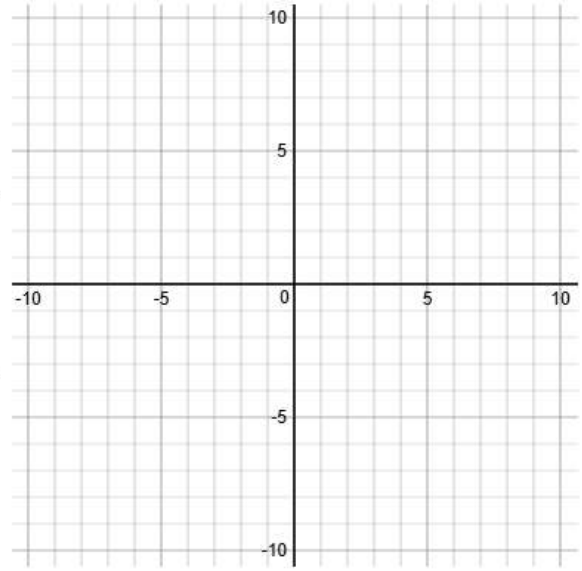
4) $x^2 + x - 30 \geq 0$

a) Graph the function $y = x^2 + x - 30$

b) The x-intercepts of the function $y = x^2 + x - 30$ are _____?_____

c) The interval on which the graph $y = x^2 + x - 30$ is above the x-axis is _____?_____

d) The solution set for the inequality $x^2 + x - 30 \geq 0$ is _____?_____



Solve each rational inequality and graph the solution set.

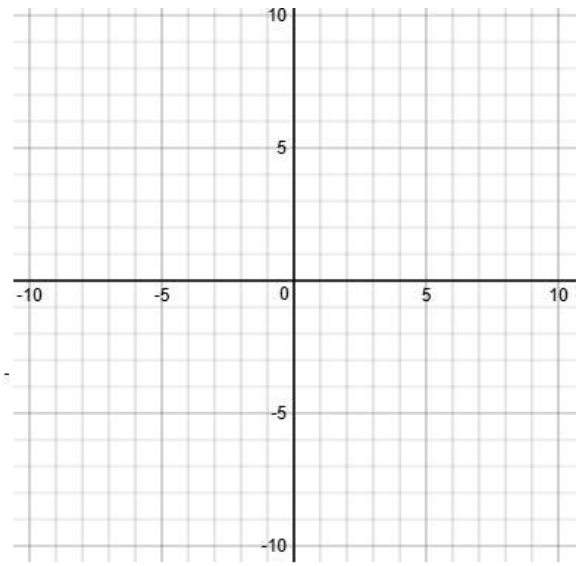
5) $4x^2 + 11x + 6 < 0$

a) Graph the function $y = 4x^2 + 11x + 6$

b) The x-intercepts of the function $y = 4x^2 + 11x + 6$ are ___?___

c) The interval on which the graph $y = 4x^2 + 11x + 6$ is below the x-axis is _____?_____

d) The solution set for the inequality $4x^2 + 11x + 6 < 0$ is ___?___



Solve each rational inequality and graph the solution set.

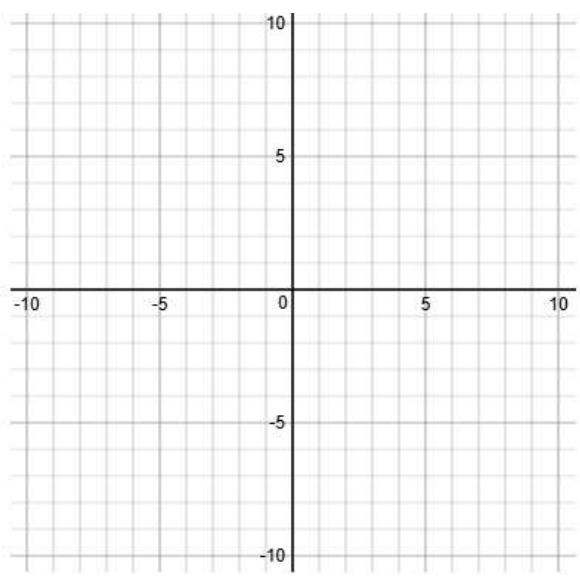
6) $2x^2 + x - 10 \leq 0$

a) Graph the function $y = 2x^2 + x - 10$

b) The x-intercepts of the function $y = 2x^2 + x - 10$ are ___?___

c) The interval on which the graph $y = 2x^2 + x - 10$ is below the x-axis is _____?_____

d) The solution set for the inequality $2x^2 + x - 10 \leq 0$ is ___?___



Solve each rational inequality and graph the solution set.

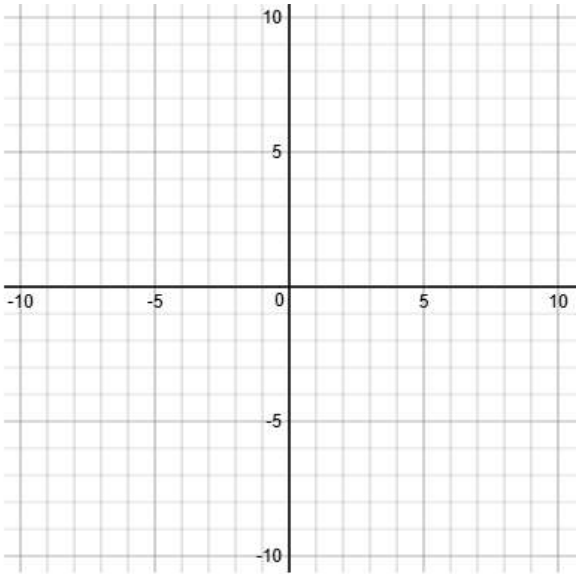
7) $4x^2 - 5x \le 6$

a) Graph the function $y = 4x^2 - 5x - 6$

b) The x-intercepts of the function $y = 4x^2 - 5x - 6$ are __?__

c) The interval on which the graph $y = 4x^2 - 5x - 6$ is below the x-axis is _____?_____

d) The solution set for the inequality $4x^2 - 5x \le 6$ is __?__



Solve each rational inequality and graph the solution set.

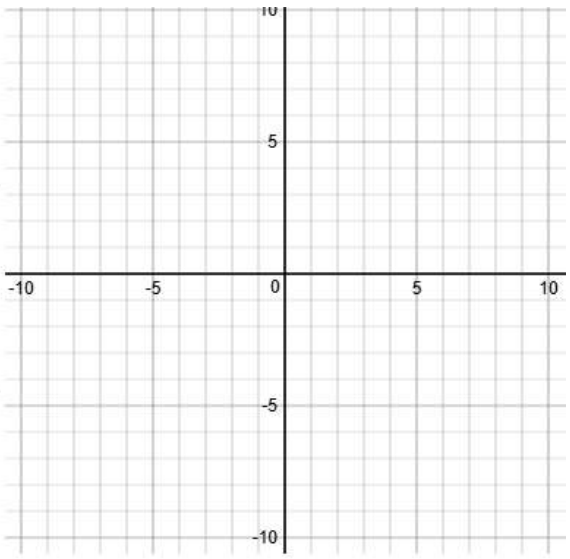
8) $x^3 + 2x^2 - 13x + 10 < 0$

a) Graph the function $y = x^3 + 2x^2 - 13x + 10$

b) The x-intercepts of the function $y = x^3 + 2x^2 - 13x + 10$ are __?__

c) The interval on which the graph $y = x^3 + 2x^2 - 13x + 10$ is below the x-axis is _____?_____

d) The solution set for the inequality $x^3 + 2x^2 - 13x + 10 < 0$ is __?__



Solve each rational inequality and graph the solution set.

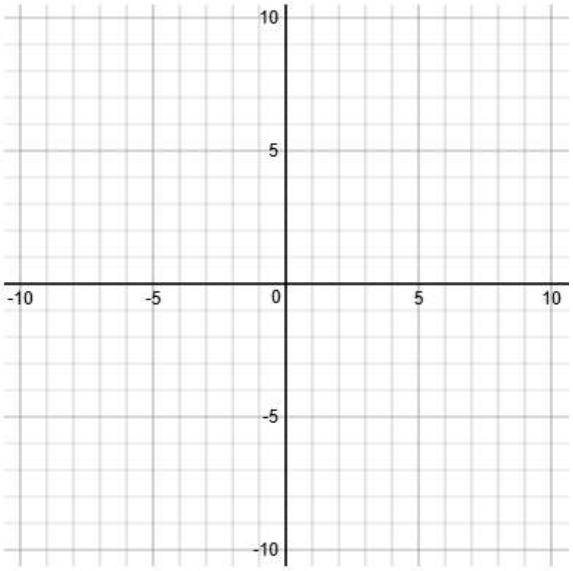
9) $x^3 - 7x + 6 < 0$

a) Graph the function $y = x^3 - 7x + 6$

b) The x-intercepts of the function $y = x^3 - 7x + 6$ are __?__

c) The interval on which the graph $y = x^3 - 7x + 6$ is below the x-axis is _____?_____

d) The solution set for the inequality $x^3 - 7x + 6 < 0$ is __?__



Solve each rational inequality and graph the solution set.

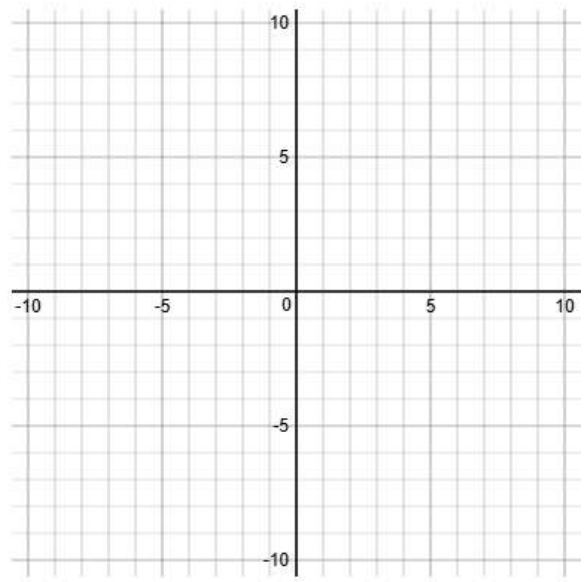
10) $x^3 \geq 6x^2$

a) Graph the function $y = x^3 - 6x^2$

b) The x-intercepts of the function $y = x^3 - 6x^2$ are ____?____

c) The interval on which the graph $y = x^3 - 6x^2$ is above the x-axis is ____?____

d) The solution set for the inequality $x^3 \geq 6x^2$ is ___?___



11) A company manufactures shoes and produce x units of pairs of shoes.

The revenue function is $R(x) = -0.5x^2 + 100x$.

The cost function is $C(x) = 20x + 1000$

The profit inequality is $R(x) > C(x)$ or $-0.5x^2 + 100x > 20x + 1000$

Solve the profit inequality $-0.5x^2 + 100x > 20x + 1000$ and graph the solution set.

a) Graph the function $y = -0.5x^2 + 100x - 20x - 1000$

b) The x-intercepts of the function $y = -0.5x^2 + 100x - 20x - 1000$ ____?____

c) The interval on which the graph $y = -0.5x^2 + 100x - 20x - 1000$ is above the x-axis is ____?____

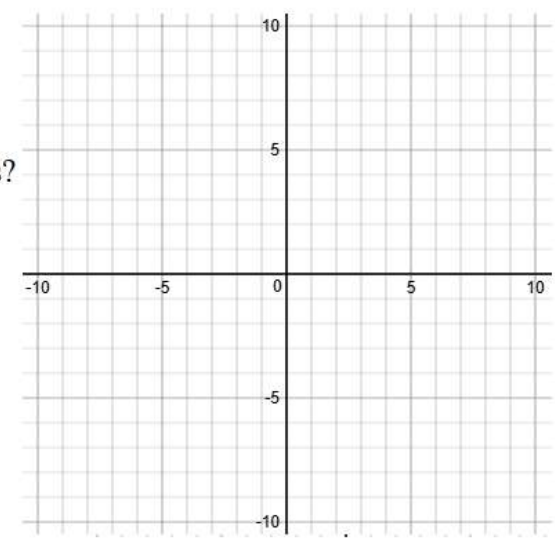
d) The solution set for the inequality $-0.5x^2 + 100x > 20x + 1000$ is ____?____

12) A ball is thrown straight up, from 2 meters above the ground, with a velocity of 12 m/s.

The height of the ball at time t is $h(t) = 2 + 12t - 5t^2$

During which time period will the ball's height exceed 6 meters?

a) Graph the function $y = 2 + 12t - 5t^2$ and $y = 6$

b) The interval on which the graph $y = 2 + 12t - 5t^2$ is above the line $y = 6$ ___?___

d) Time period during which ball exceeds 6 meters is ____?____